

TASK_1

Command History — Up/Down arrow, escape sequences, input buffer.

```
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ nano history.c
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ gcc history.c -o history
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ ./history
myshell> hello
Command: hello
myshell> ls
Command: ls
myshell> pwd
Command: pwd
myshell> exit

Exiting...
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ |
```

PART 2 — Dynamic Memory Management

```
Exiting...
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ nano memory.c
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ gcc memory.c -o memory
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ ./memory
Enter initial array size: 3
Enter 3 numbers:
10 20 30
Enter new array size: 5
Enter 2 additional numbers:
40 50

Array elements:
10 20 30 40 50
Linked list created successfully.
Node data: First Node
Memory released successfully.
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ |
```

PART 3 — Install Valgrind

```
Memory released successfully.
sairedy@DESKTOP-HKQRL9G:~/OSSP/skill$ sudo apt update
[sudo] password for sairedy:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease
[126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [
126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd6
4 Components [46.3 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
[126 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/universe
amd64 Components [76.2 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64
Packages [1187 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main Transl
ation-en [281 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64
Components [180 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/universe a
md64 Packages [1682 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/universe a
md64 Components [388 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/restricted
amd64 Packages [1416 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/restricted
Translation-en [321 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/multiverse
amd64 Components [940 B]
Get:15 http://archive.ubuntu.com/ubuntu noble-backports/main amd
64 Components [5776 B]
Get:16 http://archive.ubuntu.com/ubuntu noble-backports/universe
amd64 Components [12.6 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/main amd
64 Components [5776 B]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/universe
amd64 Components [12.6 kB]
```

```
Reading repository data... Done
Reading state information... Done
71 packages can be upgraded. Run 'apt list --upgradable' to see them.
[sudo] password for sairedy:
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  gdb libbabeltrace1 libc-dbg libdebuginfod-common
  libdebuginfod-tls libipt2 libsource-highlight-common
  libsource-highlight-tls
Suggested packages:
  gdb-doc glibc-source valgrind-dbg valgrind-mpi kcachegrind
  allegro valgrind
The following NEW packages will be installed:
  gdb libbabeltrace1 libc-dbg libdebuginfod-common
  libdebuginfod-tls libipt2 libsource-highlight-common
  libsource-highlight-tls valgrind
0 upgraded, 9 newly installed, 0 to remove and 71 not upgraded.
Need to get 27.8 MB of archives.
After this operation, 183 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libdebuginfod-common all 0.190-1.1ubuntu1 [14.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libbabeltrace1 amd64 1.5.11-3build1 [164 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libdebuginfod-tls amd64 0.190-1.1ubuntu1 [17.1 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 libipt2 amd64 2.0.6-1build1 [45.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 libsource-highlight-common all 3.1.9-4.3build1 [64.2 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 libsource-highlight-tls amd64 3.1.9-4.3build1 [258 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gdb amd64 15.1-1ubuntu1~24.04 [2402 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libc-dbg amd64 2.39-0ubuntu8.8 [7981 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/main amd64 valgrind amd64 3.22.0-0ubuntu1 [14.9 MB]
Fetched 27.8 MB in 10s (2601 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libdebuginfod-common.
(Reading database ... 46694 files and directories currently installed.)
Preparing to unpack .../9-libdebuginfod-common.0.190-1.1ubuntu1.1.all.deb ...
Unpacking libdebuginfod-common (0.190-1.1ubuntu1) ...
Selecting previously unselected package libbabeltrace1:amd64.
Preparing to unpack .../7-libbabeltrace1.1.5.11-3build1.amd64.deb ...
Unpacking libbabeltrace1:amd64 (1.5.11-3build1) ...
Selecting previously unselected package libdebuginfod-tls:amd64.
Preparing to unpack .../2-libdebuginfod-tls.0.190-1.1ubuntu1.1.amd64.deb ...
Unpacking libdebuginfod-tls:amd64 (0.190-1.1ubuntu1) ...
Selecting previously unselected package libipt2.
Preparing to unpack .../3-libipt2.2.0.6-1build1.amd64.deb ...
Unpacking libipt2 (2.0.6-1build1) ...
Selecting previously unselected package libsource-highlight-common.
Preparing to unpack .../6-libsource-highlight-common.3.1.9-4.3build1.all.deb ...
Unpacking libsource-highlight-common (3.1.9-4.3build1) ...
Selecting previously unselected package libsource-highlight-tls:amd64.
Preparing to unpack .../5-libsource-highlight-tls.3.1.9-4.3build1.amd64.deb ...
Unpacking libsource-highlight-tls:amd64 (3.1.9-4.3build1) ...
Selecting previously unselected package gdb.
Preparing to unpack .../6-gdb.15.1-1ubuntu1~24.04.1.amd64.deb ...
Unpacking gdb (15.1-1ubuntu1~24.04.1) ...
Selecting previously unselected package libc-dbg:amd64.
Preparing to unpack .../7-libc-dbg.2.39-0ubuntu8.8.amd64.deb ...
Unpacking libc-dbg:amd64 (2.39-0ubuntu8) ...
Selecting previously unselected package valgrind.
Preparing to unpack .../8-valgrind.3.22.0-0ubuntu1.amd64.deb ...
Unpacking valgrind (3.22.0-0ubuntu1) ...
Setting up libdebuginfod-common (0.190-1.1ubuntu1) ...
Setting up libsource-highlight-common (3.1.9-4.3build1) ...
Setting up libipt2 (2.0.6-1build1) ...
Setting up libbabeltrace1:amd64 (1.5.11-3build1) ...
Setting up valgrind (3.22.0-0ubuntu1) ...
Setting up libdebuginfod-tls:amd64 (0.190-1.1ubuntu1) ...
Setting up libsource-highlight-tls:amd64 (3.1.9-4.3build1) ...
Setting up gdb (15.1-1ubuntu1~24.04.1) ...
Processing triggers for man-db (2.12.0-3build1) ...
Processing triggers for libc-bin (2.39-0ubuntu8) ...
```

```

Processing triggers for libc-bin (2.39-0ubuntu8.8) ...
sairedy@DESKTOP-HKQRL9G:~/OSSP/skill$ valgrind ./memory
==1059== Memcheck, a memory error detector
==1059== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==1059== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==1059== Command: ./memory
==1059==
Enter initial array size: 3
Enter 3 numbers:
10 20 30
Enter new array size: 5
Enter 2 additional numbers:
40 50

Array elements:
10 20 30 40 50
Linked list created successfully.
Node data: First Node
Memory released successfully.
==1059==
==1059== HEAP SUMMARY:
==1059==    in use at exit: 0 bytes in 0 blocks
==1059==   total heap usage: 5 allocs, 5 frees, 2,144 bytes allocated
==1059==
==1059== All heap blocks were freed -- no leaks are possible
==1059==
==1059== For lists of detected and suppressed errors, rerun with: -s
==1059== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
sairedy@DESKTOP-HKQRL9G:~/OSSP/skill$

```

Memory Leak

Memory that was allocated but never released using free().

Buffer Overflow

When a program writes beyond the allocated size of a buffer.

Valgrind

A tool used to detect memory problems such as:

- Memory leaks
- Invalid memory access
- Incorrect use of allocated memory

TASK 1 — Tokenization / Lexical Analysis

```
==1059== All heap blocks were freed -- no leaks are possible
==1059==
==1059== For lists of detected and suppressed errors, rerun with: -s
==1059== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ nano lexer.c
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ gcc lexer.c -o lexer
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ ./lexer
Enter a command: ls -l file.txt

Tokens:
Token 1: ls
Token 2: -l
Token 3: file.txt

Total tokens: 3
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ |
```

TASK 2 — Parser Logic / Parse Tree

```
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ nano lexer.c
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ gcc lexer.c -o lexer
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ ./lexer
Enter a command: ls -l file.txt

Tokens:
Token 1: ls
Token 2: -l
Token 3: file.txt

Total tokens: 3
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ nano parser.c
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ gcc parser.c -o parser
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ ./parser
Enter command: ls

Parse Result:
Command: ls
Argument: None

Parse Tree:
Command
ls

Syntax is valid.
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ |
```

Test an empty command

```
Syntax is valid.
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ ./parser
Enter command:
Error: Empty command
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ |
```

Test a command without argument

```
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$ ./parser
Enter command: pwd

Parse Result:
Command: pwd
Argument: None

Parse Tree:
Command
|
pwd

Syntax is valid.
saireddy@DESKTOP-HKQRL9G:~/OSSP/skill$
```